

Stats for Sociologists

Course: SOC6302H1F

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Form Code: A

INSTRUCTIONS:

Fill out the first page of the scantron. Confirm you have included:

- The form code (A or B)
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1. What will happen if you try to use a function from a package that hasn't been loaded?
 - (a) R will automatically load the package
 - (b) The function will run normally
 - (c) R will prompt you to install the package
 - (d) You will get an error saying the function is not found
2. According to Wheelan, what is the most common cause of inaccurate polling results?
 - (a) Misuse of statistical software during data analysis
 - (b) Incorrect calculations of standard errors
 - (c) Random variation in respondent answers
 - (d) Biased sampling methods or poorly worded questions
3. Using the U.S. General Social Survey data, a researcher tests whether the proportions who favor a law requiring a person to obtain a police permit before they could buy a gun have changed over time. Assuming an α level of .05, which of these statements best summarizes the researcher's conclusion?

t-statistic	df	p-value	95% CI Lower	95% CI Upper
1.8	3115	0.07	0	0.06

- (a) There is not a statistically significant difference in support for the police permit law between 1984 and 2024, indicating that public opinion has remained stable.
 - (b) There is a statistically significant increase in support for the police permit law from 1984 to 2024, proving that stricter gun control laws are now favored by most Americans.
 - (c) The difference in support for the police permit law between 1984 and 2024 was due to sampling error and cannot be interpreted.
 - (d) There is a statistically significant difference in support for the police permit law between 1984 and 2024, suggesting that public opinion has changed over time.
4. Researchers calculated a 95% confidence interval of the average family size of Canadians to be from 2.7 to 3.1, based on a sample of 500 families with a standard deviation of 1.1. What change would produce a more precise estimate?
 - (a) Increasing our confidence interval to 99%
 - (b) Improving our sampling strategy
 - (c) Increasing our standard deviation to 1.3
 - (d) Increasing our sample size to 600
5. Which function is used to load a package in R?
 - (a) `load()`
 - (b) `source()`
 - (c) `install.packages()`
 - (d) `library()`
6. A researcher compares support for universal healthcare (Support vs. Oppose) between two racial groups: Black and White respondents. Which statistical test should be used?
 - (a) Pearson's correlation (r)
 - (b) Chi-square test
 - (c) Linear regression
 - (d) t-test
7. Which command installs a new package in R?
 - (a) `load()`
 - (b) `require()`
 - (c) `library()`
 - (d) `install.packages()`

8. Why is it considered good practice to use RStudio Projects when working on data analysis?
- (a) They prevent syntax errors by enforcing tidyverse formatting rules
 - (b) They automatically load packages when you open RStudio, so there's no need to use `library()`
 - (c) They ensure your code runs faster by optimizing memory usage
 - (d) They help organize your files, set a working directory, and keep your environment consistent

9. What does the following R code produce?

```
my_data |>
  tbl_cross(
    row = incgap,
    col = sex,
    percent = "column",
    missing = "no") |>
  add_p(source_note = TRUE)
```

- (a) A summary table comparing the mean values of `incgap` between men and women using a t-test
 - (b) A regression output predicting `incgap` from `sex`, including R-squared and p-values
 - (c) A frequency table of `incgap` responses with missing values imputed and grouped by `sex`
 - (d) A cross-tabulation of `incgap` by `sex`, showing column percentages and a chi-square test of independence
10. Based on the table below, which statement best compares the relative frequency of being `not too happy` between men and women?

	sex	male	female	Total
happy				
very happy		293 (20.2%)	386 (21.3%)	679 (20.8%)
pretty happy		830 (57.1%)	1054 (58.2%)	1884 (57.7%)
not too happy		330 (22.7%)	370 (20.4%)	700 (21.5%)
Total		1453 (100.0%)	1810 (100.0%)	3263 (100.0%)

- (a) A greater number of women reported being `not too happy` than men, so women have a higher relative frequency.
 - (b) The frequency of being `not too happy` is equal for men and women.
 - (c) Men have a slightly higher relative frequency of being `not too happy` than women, even though fewer men reported it.
 - (d) Women have a slightly higher relative frequency because more women than men reported being `not too happy`.
11. According to Cohen's argument, what role should descriptive statistics play in social science research?
- (a) Descriptive statistics are useful only as a predominately as a step before hypothesis testing and theory development.
 - (b) Descriptive work should be minimized in favor of studies that make strong theoretical contributions.
 - (c) Descriptive statistics are a foundational part of scientific inquiry and should be valued as legitimate research.
 - (d) Descriptive statistics are outdated and should be replaced by more advanced inferential methods.
12. In a normally distributed set of test scores, a student scoring below the average will have a _____.
- (a) positive z-score
 - (b) positive standard error
 - (c) negative standard error
 - (d) negative z-score

t-statistic	df	p-value	95% CI Lower	95% CI Upper
-2.8	3617	0	-0.43	-0.08

13. Using the U.S. General Social Survey data, a researcher suspects Americans are watching more tv today than in the 1970s. Assuming an α level of .05, should the researcher reject the null hypothesis? Why or why not?

- (a) No, because the p-value (0.005) equals the α level of .05.
- (b) No, because the p-value (0.005) is greater than the α level of .05.
- (c) Yes, because the p-value (0.005) is greater than the α level of .05.
- (d) Yes, because the p-value (0.005) is less than the α level of .05.

14. A p-value is

- (a) A measure that tells us the variability in a normal distribution
- (b) A measure of how unusual or rare our obtained statistic is compared with what is stated in the null hypothesis
- (c) An estimate of the standard error of a sampling distribution
- (d) An estimate of where our population statistic lies on the true sampling distribution of the means

15. The level of measurement for the `hapmar` variable is _____

[1] "Happiness of marriage"

	Freq	%	% Cum.
very happy	802	59.36	59.36
pretty happy	504	37.31	96.67
not too happy	45	3.33	100.00
Total	1351	100.00	100.00

- (a) dichotomous
- (b) interval
- (c) ordinal
- (d) nominal

16. The U.S. General Social Survey asked people how many brothers and sisters they had. A researcher used R to produce the summary statistics of this variable (`sibs`) below. Based on this, you conclude the distribution for this variable is:

freq	median	mean	sd
3291	3	2.9	1.8

- (a) positively skewed
- (b) negatively skewed
- (c) nominal
- (d) symmetrical

17. Which of the following code snippets correctly uses `mutate()` to convert height from centimeters to meters?

- (a) `mutate(height / 100 = height)`
- (b) `mutate(height = height / 100)`
- (c) `mutate(height = select(height / 100))`
- (d) `mutate(height <- height / 100)`

18. What is Wheelan's primary message of Chapter 1 ("What's the Point?") in *Naked Statistics: Stripping the Dread from the Data*?

- (a) Statistics provide honest answers to complex questions
- (b) Statistical formulas are the most important part of understanding data
- (c) Statistics explain why relationship exist between variables
- (d) Statistics simplify complex information to help³ us make better decisions

19. National survey data shows that 69% of people support mandatory childhood vaccinations. A local researcher believes that support is likely much higher in Toronto compared with the national average. Which of the following equations represents the research hypothesis?
- $\mu < 0.69$
 - $\mu > 0.69$
 - $\mu \neq 0.69$
 - $\mu = 0.69$
20. Which of the following is a measure of the spread of a distribution?
- mean
 - standard deviation
 - median
 - standardized variable
21. Which of these datasets has a standard deviation of 0?
- 4, 4, 4, 4, 4
 - 2, 2, 5, 7, 7
 - 1, 4, 7, 10, 15
 - 0, 2, 4, 6, 8
22. A sociologist wants to examine whether there is a relationship between level of education (measured in years) and annual income (in dollars) among adults in the U.S. Which statistical test is most appropriate?
- Chi-square test
 - t-test
 - Logistic regression
 - Pearson's correlation (r)
23. Chi-square values cannot be ____.
- outside the range of 0 to 1
 - zero
 - negative
 - positive
24. Which of the following is a valid use of `group_by()` and `summarize()`?
- `group_by(data, summarize(sex))`
 - `data |> group_by(sex) |> summarize(freq = n())`
 - `summarize(group_by(data, age), mean(sex))`
 - `data |> summarize(mean = group_by(sex))`
25. Which description best describes the correlation between personal income (`realinc`) and number of days of poor mental health (`mntlhlth`)?

Correlation Coefficient	t-statistic	df	p-value	95% CI Lower	95% CI Upper
-0.11	-5	1933	0	-0.16	-0.07

- There is a weakly positive association between personal income and days of poor mental health ($r = -0.11$, $p < .05$).
- Results indicated an inverse relationship between personal income and days of poor mental health ($r = -0.11$, $p < .05$), suggesting that the number of days of poor mental health decreases as personal income increases.
- There is no evidence of an association between personal income and days of poor mental health ($t = -5$).
- Decreases in personal income was correlated with decreases in the number of days of poor mental health ($t = -5$, $p < .05$).

26. What does the `::` operator do in R?
- (a) Assigns a value to a variable
 - (b) Comments out a line of code
 - (c) Specifies a function from a particular package (without loading it)
 - (d) Imports all functions from a package
27. According to the 68-95-99.7 rule, what can we say about a person whose score is more than 2 standard deviations above the mean?
- (a) Their score is very common.
 - (b) Their score is likely below average.
 - (c) Their score is unusual and higher than most.
 - (d) Their score is within the middle 68% of the distribution.
28. The variable `mntlhlth` measures how many days in the past 30 a respondent reported poor mental health. A respondent reports 5 poor mental health days. Based on the summary statistics for this variable, which of the following best describes their z-score?

freq	median	mean	sd
2279	0	4.5	7.9

- (a) The z-score is close to zero because 5 is close to the mean.
 - (b) The z-score is negative because 5 is below the sd.
 - (c) The z-score is large and positive because 5 is much higher than the mean.
 - (d) The z-score is undefined because the median is 0.
29. What key concept does Wheelan illustrate in the chapter titled Inference: Why My Statistics Professor Thought I Might Have Cheated?
- (a) That cheating can be detected by comparing mean scores across different groups
 - (b) That hypothesis testing is not useful in criminal investigations
 - (c) That statistical inference can be used to prove someone's guilt beyond reasonable doubt
 - (d) That unusually accurate results can raise suspicion, but statistical inference helps assess whether they are likely due to chance
30. Which description best describes a correlation coefficient of $r = -0.62$?
- (a) Weakly negative relationship
 - (b) A statistically insignificant relationship
 - (c) Strong negative relationship
 - (d) No relationship exists between the variables
31. If a researcher concluded that there was no difference in police use of force against Black and White men when there actually is a difference, then:
- (a) The null hypothesis was correctly rejected.
 - (b) The null hypothesis was correctly retained.
 - (c) The null hypothesis was incorrectly rejected (false positive).
 - (d) The null hypothesis was incorrectly retained (false negative).
32. Which RStudio pane is used to write and edit R scripts and Quarto documents?
- (a) Source
 - (b) Environment
 - (c) Files
 - (d) Console

33. What does the pipe operator (`|>`) do in `dplyr`?
- (a) It loads the `dplyr` package into memory.
 - (b) It filters rows based on a condition.
 - (c) It passes the result of one function into the next.
 - (d) It creates a new column in a data frame.
34. Which of the following best describes the differences between the Console, R Scripts, and Quarto documents in RStudio?
- (a) R Scripts and Quarto documents both run code line-by-line, but only the Console saves your work.
 - (b) The Console and R Scripts are identical, while Quarto is used only for Markdown formatting.
 - (c) The Console is used for writing long-term code, R Scripts are for temporary testing, and Quarto is only for plotting.
 - (d) The Console runs code interactively, R Scripts store reusable code, and Quarto combines code, text, and output for dynamic reports.
35. A researcher used the 2024 General Social Survey to find the proportion of the U.S. population that reports believing in some form of life after death. Based on the R output with 90% confidence intervals, which of the following is a correct interpretation of their results?

n	mean	sd	lowCI	hiCI
1283	0.83	0.38	0.81	0.85

- (a) About 90% of people responded to the national survey.
 - (b) If another survey had been conducted in 2024, there's a 90% chance the point estimate would be between .81 and .85.
 - (c) The researcher is 90% confident that the proportion of U.S. adults who believe in some form of life after death is between .81 and .85.
 - (d) Because the mean is within the confidence interval, there is a .90 probability that the proportion of U.S. adults who believe in some form of life after death is .83.
36. What is the purpose of `drop_na()` in `dplyr`?
- (a) It drops columns that are entirely NA.
 - (b) It removes rows that contain missing values.
 - (c) It replaces missing values with zeros.
 - (d) It converts NA values to NULL.
37. What will the following code do?
- ```
gss24 <- gss24 |>
 mutate(status = case_when(
 income > 100000 ~ "High",
 income > 50000 ~ "Medium",
 TRUE ~ "Low"
))
```
- (a) It removes missing values from the income column.
  - (b) It replaces the numeric values in the income column with text labels.
  - (c) It creates a new column `status` with 3 possible values.
  - (d) It filters out rows where income is less than 50,000.

38. When examining a frequency distribution, which of the following columns generally allows you to identify the median most quickly?
- percentages
  - frequencies
  - cumulative frequencies
  - cumulative percentages
39. In Naked Statistics, Charles Wheelan explains that “most sample means will lie reasonably close to the population mean; the standard error is what defines ‘reasonably close.’” What does this statement imply?
- The standard error tells us how far the population mean is from the sample mean.
  - The sample mean is always equal to the population mean.
  - The standard error measures how much sample means tend to vary around the population mean.
  - The standard error increases as sample size increases.
40. What does Oscar Gandy suggest about the use of racial statistics?
- Their use in data journalism helps mitigate online polarization
  - Their use in legal proceedings increasingly lead to unfair policy outcomes
  - They can both expose and perpetuate discrimination
  - They are neutral tools most useful for academic research
41. Which of the following statements best reflects the ASA’s guidance on interpreting p-values?
- A large p-value proves that no effect exists.
  - P-values measure the scientific or economic significance of a result.
  - P-values are influenced by sample size and measurement precision, and do not reflect effect size or importance.
  - A small p-value always indicates a large and important effect.
42. Has public opinion changed about whether Black Americans have worse jobs, income, and housing than white Americans mainly because of discrimination? Based on a significance level of  $\alpha = 0.01$ , what’s the best conclusion from the test?

|         | year         |              | Total        |
|---------|--------------|--------------|--------------|
|         | 2022         | 2024         |              |
| opinion |              |              |              |
| no      | 509 (45%)    | 522 (51%)    | 1,031 (48%)  |
| yes     | 619 (55%)    | 494 (49%)    | 1,113 (52%)  |
| Total   | 1,128 (100%) | 1,016 (100%) | 2,144 (100%) |

Pearson’s Chi-squared test,  $p=0.004$

- The evidence shows attitudes about racial inequality differ between 2022 and 2024.
  - There is no statistically significant difference in attitudes between 2022 and 2024.
  - The p-value of .000 means the null hypothesis is true.
  - The results show that most people in both years agreed racial inequality is due to discrimination.
43. In order to compare meaningfully across groups, a frequency should be converted into a/an \_\_\_\_\_.
- number
  - ordinal variable
  - standard deviation
  - percentage
44. A researcher used the 2024 General Social Survey to find the proportion of the U.S. population that reports believing in some form of life after death. Based on the R output of their findings, with 95% confidence intervals, which of the following is a correct interpretation of their results?



| sex    | n    | mean | sd  | lowCI | hiCI |
|--------|------|------|-----|-------|------|
| male   | 1139 | 4.2  | 3.7 | 4.0   | 4.5  |
| female | 1174 | 4.0  | 3.5 | 3.8   | 4.2  |

- (a) We cannot determine whether the null hypothesis of zero difference between the parameters will be rejected at the 0.05 level.
  - (b) The confidence intervals do not overlap so we can conclude that the average time men and women spend on leisure activities per day are statistically significantly different.
  - (c) We are 95% certain that women have less time per day than men for leisure activities, likely because of their greater time spent on housework and childcare.
  - (d) The confidence intervals overlap, so we can't know for sure if there is a statistical difference between the average time men and women spend on leisure activities per day.
45. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?
- (a) This couple had about 8 arguments per month.
  - (b) This couple had fewer arguments than about 2% of couples.
  - (c) This couple had fewer arguments than about 98% of couples.
  - (d) This couple had fewer arguments than about 5% of couples.
46. What does the following code return?
- ```
gss24 |>
  group_by(sex) |>
  summarize(mean_age = mean(age, na.rm = TRUE))
```
- (a) A new column named mean_age added to the original data frame.
 - (b) a data frame showing the average age for each sex.
 - (c) a list of all ages grouped by sex.
 - (d) A filtered data frame with only rows where age is not missing.
47. What is the result of the following code?
- ```
gss24 |>
 filter(age > 20) |>
 select(age, childs)
```
- (a) Only rows with age > 20 and columns age and childs are returned.
  - (b) A new column is added to gss24
  - (c) The original gss24 dataset is overwritten.
  - (d) rows with age > 20 are removed
48. If a statistics classes' final grades were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?
- (a) 68%
  - (b) 95%
  - (c) 50%
  - (d) 62%
49. What does the expression `5 != 3` evaluate to in R?
- (a) TRUE
  - (b) NA
  - (c) NULL
  - (d) FALSE

50. Using the table, which of the following is the correct formula to find the expected frequency for U.S. liberals who oppose the death penalty in 2024?

|        | ideology | conservative | liberal     | Total         |
|--------|----------|--------------|-------------|---------------|
| cappun |          |              |             |               |
| favor  |          | 512 (66.8%)  | 255 (33.2%) | 767 (100.0%)  |
| oppose |          | 168 (32.7%)  | 345 (67.3%) | 513 (100.0%)  |
| Total  |          | 680 (53.1%)  | 600 (46.9%) | 1280 (100.0%) |

- (a)  $(600 * 767)/1280$
- (b)  $(345 * 600)/1280$
- (c)  $(345 * 513)/1280$
- (d)  $(600 * 513)/1280$